PARIKSHA365 — STUDY NOTES

Environment & Ecology — Common Book

SSC, RRB, State PSC — Biomes, Climate, Acts, Protected Areas

SSC RRB STATE PSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

Examiner's Blueprint — What Environment Questions Are Really About

⚠ EXAM ALERT

Environment & Ecology is one of the most rewarding GA topics if you study it systematically — because examiners repeat the same themes year after year. Here is how they think: **1. Conventions and treaties with years** — "Ramsar Convention year?", "When was the Montreal Protocol?", "What is COP-28 known for?" — these appear in every SSC CGL paper. **2. Protected areas — the "first" and "only" facts** — India's first National Park, the only floating NP, the only Asiatic lion habitat, the latest tiger reserve. Examiners love these landmarks. **3. National symbols and conservation projects** — National animal, aquatic animal, heritage animal; Project Tiger vs Project Elephant vs Project Cheetah. Clean 1-2 marks every paper. **4. Laws and Acts with years** — EPA 1986, Wildlife Protection Act 1972, Forest Rights Act 2006, NGT Act 2010. The year is almost always part of the question. **5. India's climate commitments** — Net-zero year (2070), non-fossil target (50% by 2030), PANCHAMRIT (5 points from Glasgow 2021). Everything else — biogeochemical cycles, IUCN categories, biodiversity hotspots, pollution parameters — appears occasionally. This book covers all of them, but focus your revision in the order listed above.

Priority order for your study time:

1. International conventions table (Part Y) — year + key fact for each
2. India's acts with years (Part F, Part X) — especially EPA 1986, WLPA 1972, NGT 2010
3. Conservation projects — Tiger, Elephant, Cheetah, Dolphin, Lion (Part Z)
4. Key protected areas — first NP, floating NP, Gir, Kaziranga, Sundarbans (Part C)
5. National symbols (Part Z)
6. Climate commitments — PANCHAMRIT, Paris Agreement, net-zero 2070 (Part E)
7. Biodiversity hotspots (Part B)
8. Pollution metrics — AQI, BOD, DO (Part D)
9. Biogeochemical cycles (Part A)
10. Ecological pyramids and energy flow (Part A)

Current counts to know (verify before any exam after 2026):

Item	Value (April 2026)
Tiger Reserves	58 (56th = Guru Ghasidas-Tamor Pingla, CG; 57th = Ratapani, MP; 58th = Madhav, MP)

Item	Value (April 2026)
Biosphere Reserves	18 in India (12 listed under UNESCO MAB)
Ramsar Sites	85+ (Tamil Nadu has the most at state level, 14+)
Tiger population	~3,167 (2022 census); 2026 census ongoing
National Parks	106
Wildlife Sanctuaries	~570

PART A — ECOLOGY FUNDAMENTALS

Chapter A1 — The Ecosystem

Every ecosystem, whether a pond or a rainforest, has the same basic structure: things that make energy, things that consume it, and things that recycle the leftovers.

- **Abiotic components** — the non-living environment: sunlight, temperature, water, air, soil, minerals.
- **Biotic components** — the living world: producers, consumers, decomposers.

Energy Flow in Food Chains

Sunlight → Producers (green plants, phytoplankton) → Primary consumers (herbivores) → Secondary consumers (carnivores) → Tertiary consumers → Decomposers (bacteria, fungi) → nutrient return.

✔ KEY FACT

The **10% Rule (Lindeman, 1942)** — only about 10% of the energy at one trophic level is transferred to the next. This is why food chains have a maximum of 4-5 levels. If you eat a plant, you get 100 units of energy. If a cow eats the plant, you get 10 units from eating the cow. This is why a vegetarian diet is more energy-efficient and why apex predators are always rare.

Ecological Pyramids

Pyramid type	Can it be inverted?	Why
Pyramid of numbers	Yes	One tree can support thousands of insects
Pyramid of biomass	Yes	In oceans, phytoplankton biomass can be less than zooplankton at any instant
Pyramid of energy	Never — always upright	Second law of thermodynamics — energy always decreases as it flows up

⚠ EXAM ALERT

"Which ecological pyramid is always upright?" — **Pyramid of energy**. This is a standard exam question. The energy pyramid can never be inverted because energy is always lost as heat at each trophic level.

Chapter A2 — Biogeochemical Cycles

Water Cycle

Evaporation from oceans → condensation into clouds → precipitation (rain/snow) → runoff into rivers → infiltration into groundwater → back to ocean. About 97% of Earth's water is in the oceans (saline). Only ~2.5% is freshwater, and most of that is locked in ice caps and glaciers.

Carbon Cycle

Plants absorb CO_2 through photosynthesis and convert it into biomass. Animals eat plants and release CO_2 through respiration. Decomposers break down dead matter, releasing CO_2. Fossil fuels are ancient biomass — burning them releases carbon that had been locked underground for millions of years, short-circuiting the natural cycle.

Nitrogen Cycle

Atmosphere is 78% N_2, but plants cannot use it directly. The cycle works through:

- **Nitrogen fixation** — converts N_2 to NH_3 usable by plants. Done by Rhizobium (in legume root nodules), Azotobacter (free-living in soil), cyanobacteria (Anabaena, Nostoc), lightning (abiotic), and the Haber-Bosch industrial process.
- **Nitrification** — bacteria convert NH_3 → NO_2^- → NO_3^- (usable by plants).
- **Denitrification** — bacteria return NO_3^- to atmospheric N_2.

⚠ EXAM ALERT

"Which element's cycle has no gaseous phase?" — **Phosphorus**. Unlike carbon, nitrogen, and sulphur, phosphorus does not enter the gaseous phase. It cycles only through rock → soil → plants → animals → decomposers → back to soil. Phosphorus is often the limiting nutrient in lakes (which is why phosphate-rich fertiliser runoff causes algal blooms).

PART B — BIODIVERSITY

Chapter B1 — Levels and Importance

- **Genetic diversity** — variation in genes within a species (e.g., different rice varieties).
- **Species diversity** — variety of species within an ecosystem.
- **Ecosystem diversity** — variety of habitats and ecosystems (forests, wetlands, grasslands).

Why Biodiversity Matters — Ecosystem Services

Category	What we get
Provisioning	Food, water, timber, medicine, raw materials
Regulating	Climate regulation, pollination, flood control, disease control
Cultural	Recreation, spiritual value, tourism, aesthetic
Supporting	Nutrient cycling, soil formation, water purification

Chapter B2 — Biodiversity Hotspots

A biodiversity hotspot must meet two strict criteria: it must have **at least 1500 species of endemic vascular plants**, AND must have lost **at least 70% of its original primary vegetation**. Norman Myers proposed the concept in 1988. There are now **36 global hotspots**.

India has 4 biodiversity hotspots:

1. **Himalayas** — Eastern Himalayan region including Bhutan, parts of Nepal and northeast India.
2. **Indo-Burma** — Northeast India, Myanmar, parts of Bangladesh, southern China.
3. **Western Ghats + Sri Lanka** — Sahyadri range + Sri Lanka.
4. **Sundaland** — India's portion is the Nicobar Islands only.

◎ MNEMONIC

India's 4 hotspots: **"Himalayas, Indo-Burma, Western Ghats, Sundaland"** — **HIWS** — "**H**ot **I**ndian **W**ildlife **S**pots"

India is also one of the **17 megadiverse countries** that together hold approximately 70% of global biodiversity. The others include Brazil, Colombia, Indonesia, Mexico, Australia, China, DRC, and more.

Chapter B3 — IUCN Red List Categories

EX (Extinct) → **EW** (Extinct in Wild) → **CR** (Critically Endangered) → **EN** (Endangered) → **VU** (Vulnerable) → **NT** (Near Threatened) → **LC** (Least Concern) → **DD** (Data Deficient) → **NE** (Not

Evaluated)

◎ **MNEMONIC**

"Every Experienced Conservationist Encourages Very Nature-Loving Decisions" — EX, EW, CR, EN, VU, NT, LC, DD (ignore NE).

IUCN (International Union for Conservation of Nature) was established in **1948** and maintains the Red List. The Great Indian Bustard is **Critically Endangered**. The Gangetic Dolphin is **Endangered**.

Chapter B4 — Key Endangered Species in India

Species	IUCN Status	Where found
Great Indian Bustard	Critically Endangered	Rajasthan (Desert NP), Gujarat
Hangul (Kashmir Stag)	Critically Endangered	Dachigam NP, J&K
Gharial	Critically Endangered	Chambal river (UP-MP-Rajasthan border)
Indian Vulture (long-billed)	Critically Endangered	Various states (crashed due to diclofenac)
Asiatic Lion	Endangered	Gir NP, Gujarat — the ONLY place in the world
Royal Bengal Tiger	Endangered	Multiple PAs; Sundarbans, Ranthambore, Kanha
Indian Elephant	Endangered	Western Ghats, northeast India
Sangai (brow-antlered deer)	Endangered	Keibul Lamjao, Manipur
Nilgiri Tahr	Endangered	Western Ghats (Eravikulam NP)
One-horned Rhinoceros	Vulnerable	Kaziranga, Manas (Assam)
Snow Leopard	Vulnerable	Hemis (Ladakh), Spiti (HP)
Olive Ridley Turtle	Vulnerable	Gahirmatha (Odisha) — world's largest nesting beach

PART C — PROTECTED AREAS AND INDIA'S WILDLIFE

Chapter C1 — Types of Protected Areas

Under the **Wildlife (Protection) Act 1972**:

1. **National Park** — highest protection; no human activity (including grazing) allowed inside. India has **106 NPs**.
2. **Wildlife Sanctuary** — lower protection than NP; some regulated activities permitted. India has **~570 WLSs**.
3. **Conservation Reserve** — on community or state land around NPs/WLSs.
4. **Community Reserve** — private or community land managed for conservation.

Tiger Reserves — administered by NTCA under Project Tiger. As of April 2026: **58 Tiger Reserves**.

Biosphere Reserves — UNESCO MAB (Man and Biosphere) programme. India has **18 BRs, of which 12 are on the UNESCO MAB World Network**.

Elephant Reserves — under Project Elephant. India has **33 Elephant Reserves**.

Chapter C2 — Key National Parks — Must-Know Facts

✔ KEY FACT

The most-tested NP facts: **India's first NP = Jim Corbett (originally Hailey NP, 1936, Uttarakhand)**. **Largest NP by area = Hemis (Ladakh, ~4,400 km²)**. **Only floating NP = Keibul Lamjao (Manipur, on Loktak Lake)**. **Only UNESCO Mixed Heritage NP = Khangchendzonga (Sikkim)**. **Only place for Asiatic Lion = Gir (Gujarat)**.

State	Key National Parks
Uttarakhand	Jim Corbett (1st NP, 1936), Nanda Devi + Valley of Flowers (UNESCO), Rajaji, Govind
Rajasthan	Ranthambore (Tiger Reserve), Sariska, Desert NP, Keoladeo Ghana (UNESCO; Siberian cranes)
Gujarat	Gir (Asiatic lion — only habitat in world), Marine NP Gulf of Kachchh (1st marine NP), Velavadar (blackbuck)
Madhya Pradesh	Kanha (barasingha; Mowgli setting), Bandhavgarh (highest tiger density in India), Pench, Panna, Satpura, Madhav
Maharashtra	Tadoba-Andhari (largest in Maharashtra), Sanjay Gandhi (Borivali), Pench
Assam	Kaziranga (UNESCO; one-horned rhinoceros), Manas (UNESCO; pygmy hog)

State	Key National Parks
Karnataka	Bandipur (1st Project Tiger reserve, 1973), Nagarhole, Kudremukh, Bannerghatta
Kerala	Silent Valley, Periyar, Eravikulam (Nilgiri tahr), Anamudi Shola
Jammu & Kashmir / Ladakh	Dachigam (Hangul deer), Hemis (largest NP; snow leopard), Salim Ali
West Bengal	Sundarbans (UNESCO; mangrove tigers), Buxa, Gorumara, Jaldapara, Singalila
Arunachal Pradesh	Namdapha, Mouling
Manipur	Keibul Lamjao (only floating NP; Sangai deer; on Loktak Lake)
Sikkim	Khangchendzonga (only UNESCO Mixed Heritage NP in India)
Himachal Pradesh	Great Himalayan NP (UNESCO), Pin Valley
Odisha	Simlipal (Biosphere Reserve + Tiger Reserve), Bhitarkanika (saltwater crocodile)

Chapter C3 — Biosphere Reserves (India's 18)

India's 18 BRs (12 are on UNESCO MAB World Network list, marked with *):

Nilgiri *(1st, 1986), Nanda Devi*, Nokrek, *Gulf of Mannar*, Sundarbans, *Manas, Similipal*, Pachmarhi, *Khangchendzonga*, Agasthyamala, *Great Nicobar*, Achanakmar-Amarkantak, *Dihang-Dibang, Dibru-Saikhowa, Panna* (listed 2020), Seshachalam, Cold Desert (HP), Kachchh (Gujarat — largest by area).

⚠ EXAM ALERT

India's first Biosphere Reserve = **Nilgiri (1986)**. Largest BR = **Kachchh, Gujarat**. Latest BR to be added to UNESCO MAB list = **Panna (2020)**. Total: 18 BRs, 12 on UNESCO list.

Chapter C4 — Ramsar Sites (Wetlands)

- **Ramsar Convention** — signed at Ramsar, Iran in **1971**; entered force 1975. India ratified in **1982**.
- India's first Ramsar sites: **Chilika (Odisha)** and **Keoladeo Ghana (Rajasthan)**, both designated in **1981** (under the 1982 ratification).
- **85+ Ramsar sites** in India as of April 2026 — highest number among Asian countries.
- **Tamil Nadu** has the most Ramsar sites at the state level (14+).
- **Largest Ramsar site in India**: Sundarbans Wetland, West Bengal.

PART D — POLLUTION

Chapter D1 — Air Pollution

Primary and Secondary Pollutants

Primary pollutants (emitted directly): CO, NO_x, SO_x, PM (particulate matter), VOCs, NH_3, heavy metals (Pb, Hg, Cd).

Secondary pollutants (formed by chemical reactions in the atmosphere): - Ground-level ozone (O_3) — formed when NO_x reacts with VOCs in sunlight. - PAN (peroxyacetyl nitrate) — component of photochemical smog. - Acid rain compounds (H_2SO_4, HNO_3) — formed when SO_2 and NO_x dissolve in rain.

✓ KEY FACT

Photochemical smog = NO_x + VOCs + sunlight → ground-level O_3 + PAN. First observed in Los Angeles. London smog (1952) was a different type — classical smog from SO_2 + coal burning + fog = "pea soup" fog. They are different problems with different causes.

Air Quality Index (India — CPCB)

AQI	Category
0–50	Good
51–100	Satisfactory
101–200	Moderate
201–300	Poor
301–400	Very Poor
401–500	Severe

AQI is maintained by CPCB (Central Pollution Control Board) and monitored for $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, NH_3, and Pb.

Chapter D2 — Water Pollution

- **BOD (Biological Oxygen Demand)** — the amount of oxygen bacteria need to decompose organic matter in water. High BOD = more organic pollution, less oxygen for fish.
- **DO (Dissolved Oxygen)** — fish need at least 4 mg/L of dissolved oxygen to survive. Polluted water can have near-zero DO.

- **Eutrophication** — excess nutrients (nitrogen, phosphorus from fertiliser runoff) → algal bloom → algae die → bacteria decompose them, consuming all oxygen → fish die. This cycle kills entire lake ecosystems.
- **Bioaccumulation** — toxic substances (DDT, Hg) build up in one organism.
- **Biomagnification** — concentration of toxic substances increases at each trophic level. Top predators (eagles, large fish) suffer the most.

Diseases from Water Pollution

Disease / Condition	Cause
Minamata disease	Mercury (Hg) poisoning — Minamata Bay, Japan (1956)
Itai-itai disease	Cadmium (Cd) poisoning — Jinzu River, Japan
Blue baby syndrome	High nitrates in water → methemoglobinemia in infants
Fluorosis	Fluoride >1.5 ppm → mottled teeth, skeletal damage
Arsenicosis	Arsenic in groundwater — severe in West Bengal and Bangladesh

Chapter D3 — Solid Waste, Noise, Soil

- **E-waste**: India generates ~3.2 million tonnes/year. E-Waste Management Rules 2022 introduced EPR.
- **Plastic**: India banned 19 categories of single-use plastics from **1 July 2022**.
- **Noise pollution**: Measured in decibels (dB). Residential area limit: 55 dB (day), 45 dB (night). Hospital/school silence zones: 50 dB (day), 40 dB (night).
- **Soil pollution**: Pesticides (DDT, organophosphates), industrial heavy metals, plastic litter, e-waste.
- **Persistent Organic Pollutants (POPs)** — DDT, dioxins. Covered by the **Stockholm Convention (2001)**.

PART E — CLIMATE CHANGE

Chapter E1 — Greenhouse Gases

The natural greenhouse effect keeps Earth ~15 °C warmer than it would otherwise be — without it, Earth's average temperature would be −18 °C. The problem is human-enhanced greenhouse warming.

Greenhouse Gas	GWP (100-year, relative to CO_2)	Main sources
CO_2	1 (reference)	Fossil fuel combustion, deforestation, cement
CH_4 (methane)	28–36	Livestock (enteric fermentation), paddy farming, landfills
N_2O	298	Fertilisers, industrial processes
HFCs	Hundreds to thousands	Refrigerants, AC units
PFCs	Thousands	Aluminium smelting
SF_6	23,500	Electrical switchgear — the strongest GHG

⚠ EXAM ALERT

The main greenhouse gas by quantity = CO_2. **The most potent greenhouse gas by GWP = SF_6** (sulphur hexafluoride, 23,500× CO_2). Examiners test both. Also: "Most abundant greenhouse gas in the atmosphere" = water vapour — but in exam context, CO_2 is the standard answer for the climate-forcing one.

Chapter E2 — Climate Treaties Timeline

Year	Event	Key Outcome
1972	Stockholm Conference on Human Environment	Birth of UNEP (HQ Nairobi); first major UN env conference
1985	Vienna Convention	Framework for ozone layer protection
1987	Montreal Protocol	Phase-out of CFCs and other ODSs; widely regarded as most successful env treaty
1988	IPCC formed	WMO + UNEP; Nobel Peace Prize 2007
1992	Rio Earth Summit (UNCED)	UNFCCC, CBD, Agenda 21 signed

Year	Event	Key Outcome
1997	Kyoto Protocol	Binding GHG cuts on developed (Annex-I) countries; India = non-Annex-I (no binding cuts)
2009	Copenhagen COP-15	Failed to achieve binding agreement; voluntary pledges only
2015	Paris Agreement (COP-21)	Limit warming to "well below 2°C, pursue 1.5°C"; voluntary NDCs by all countries
2016	Kigali Amendment (to Montreal)	Phase-down of HFCs
2021	Glasgow COP-26	India's PANCHAMRIT; coal phase-down language
2022	Sharm el-Sheikh COP-27	Loss and Damage fund agreed
2022	Kunming-Montreal GBF (CBD COP-15)	"30×30" — protect 30% of land + sea by 2030
2023	Dubai COP-28	First Global Stocktake; "transition away from fossil fuels"
2024	Baku COP-29	$300 billion/year climate finance pledge by 2035
2025	Belém COP-30	Brazil host; first COP in the Amazon

× COMMON TRAP

Common traps: Vienna Convention (1985) vs Montreal Protocol (1987) — Vienna is the framework, Montreal is the actual action. Kyoto Protocol (1997) — binding only on developed countries. Paris Agreement (2015) — voluntary NDCs for all. Stockholm Convention (2001, about POPs) is separate from Stockholm Conference (1972, about human environment). These are four different things that examiners deliberately mix up.

Chapter E3 — India's Climate Action

India's PANCHAMRIT (Glasgow COP-26, 2021) — 5 Commitments

1. Achieve **500 GW non-fossil installed capacity** by 2030.
2. **50% of energy needs** from renewables by 2030.
3. Reduce projected carbon emissions by **1 billion tonnes** by 2030.
4. Reduce **emission intensity of GDP by 45%** from 2005 level by 2030.
5. **Net-zero by 2070**.

✔ KEY FACT

India's net-zero target is **2070** — not 2050 or 2060 like most developed countries. This is important for exams. The logic: India is still developing and needs fossil fuels during its growth phase; 2070 gives 20-40 extra years compared to developed nations.

NAPCC — National Action Plan on Climate Change (2008) — 8 Missions

1. National Solar Mission
2. National Mission for Enhanced Energy Efficiency
3. National Mission on Sustainable Habitat
4. National Water Mission
5. National Mission for Sustaining the Himalayan Ecosystem
6. National Mission for a Green India
7. National Mission for Sustainable Agriculture
8. National Mission on Strategic Knowledge for Climate Change

⚠ EXAM ALERT

NAPCC has exactly **8 missions**. Examiners often ask the count. The Solar Mission (1st) and Green India Mission (6th) are the most frequently referenced.

Other Key Climate Initiatives

- **International Solar Alliance (ISA)** — launched 2015 by India and France at Paris COP-21; HQ at Gurugram, India.
- **Coalition for Disaster Resilient Infrastructure (CDRI)** — launched by India at UN Climate Action Summit, New York, 2019.
- **Mission LiFE (Lifestyle for Environment)** — launched 2022 at COP-26 by PM Modi; focuses on individual behaviour change.
- **Green Hydrogen Mission (2023)** — target: 5 million metric tonnes of green hydrogen by 2030.
- **FAME-II** — Faster Adoption and Manufacturing of Electric Vehicles.

PART F — INDIAN ENVIRONMENTAL LAWS AND INSTITUTIONS

Chapter F1 — Key Environmental Acts (Chronological)

Act	Year	What it does
Indian Forest Act	1927	British-era; defines reserved, protected, village forests
Wildlife (Protection) Act	1972	Framework for protected areas; originally 6 schedules, now 4 after 2022 amendment
Water (Prevention & Control of Pollution) Act	1974	Created CPCB and SPCBs
Forest (Conservation) Act	1980	Diversion of forest land for non-forest purposes requires Central Government approval
Air (Prevention & Control of Pollution) Act	1981	Extended CPCB's mandate to air quality
Environment (Protection) Act	**1986**	"Umbrella Act" — passed after Bhopal gas tragedy; empowers Centre to issue all environment-related rules
Public Liability Insurance Act	1991	No-fault compensation for industrial accidents involving hazardous substances
Biological Diversity Act	2002	Three-tier system: NBA + SBBs + BMCs
Forest Rights Act	2006	Rights of Scheduled Tribes and Other Traditional Forest Dwellers
National Green Tribunal (NGT) Act	**2010**	Specialised quasi-judicial body for fast disposal of environmental cases; principal bench in Delhi
Compensatory Afforestation Fund (CAMPA) Act	2016	Manages funds for afforestation when forests are diverted
Wildlife Protection Amendment Act	2022	Aligned with CITES; reduced schedules from 6 to 4
Forest Conservation Amendment Act	2023	Eased clearance for strategic/border projects

× COMMON TRAP

The **Environment Protection Act is from 1986**, NOT 1984 (the year of the Bhopal tragedy). The tragedy caused the government to pass the Act in 1986 — two years later. This distinction trips up many students.

Chapter F2 — Wildlife Protection Act 1972 — Schedules

After the 2022 Amendment, WLPA has **4 schedules** (reduced from the original 6):

- **Schedule I** — highest protection; hunting penalty up to 7 years + heavy fine. Includes tiger, lion, leopard, elephant, Great Indian Bustard, Indian rhinoceros.
- **Schedule II** — protected species with lower penalties.
- **Schedule III** — protected plants.
- **Schedule IV** — CITES-listed species (invasive alien species also included after 2022).

Chapter F3 — Key Institutions

Institution	Year	Role
CPCB	1974	Central Pollution Control Board; monitors and controls air + water pollution
MoEFCC	—	Ministry of Environment, Forest & Climate Change — apex authority
NBA	2003	National Biodiversity Authority; HQ Chennai
NTCA	2005	National Tiger Conservation Authority; Project Tiger administration
WII	1982	Wildlife Institute of India; HQ Dehradun
ZSI	1916	Zoological Survey of India; HQ Kolkata
BSI	1890	Botanical Survey of India; HQ Kolkata
FSI	1981	Forest Survey of India; publishes India State of Forest Report (biennial); HQ Dehradun
NGT	2010	National Green Tribunal; 5 benches (Delhi principal + 4 regional)
NMCG	2011	National Mission for Clean Ganga; implements Namami Gange Programme
WCCB	2007	Wildlife Crime Control Bureau — enforces wildlife laws, busts poaching networks

India State of Forest Report (ISFR) — Key Data

- Published by FSI **every 2 years**.
- Total forest + tree cover = **25.17% of geographical area** (target under National Forest Policy 1988 = 33%).
- Forest cover alone = 21.76% (~7.16 lakh km²).
- **Largest forest cover (area): MP > Arunachal > Chhattisgarh > Odisha > Maharashtra**.

- **Highest forest cover (% of state area): Mizoram > Lakshadweep > A&N > Meghalaya > Arunachal**.
- Mangrove cover: ~4,992 km². Gujarat > West Bengal > A&N (by area).

PART X — CONSERVATION PROJECTS AND NATIONAL SYMBOLS

Conservation Projects

Project	Year	Focus	Current count
Project Tiger	1973	Tiger conservation; NTCA manages	58 Tiger Reserves (April 2026)
Crocodile Conservation	1975	Gharial, mugger, saltwater crocodile	—
Project Elephant	1992	Asian elephant conservation	33 Elephant Reserves
Project Snow Leopard	2009	Cold-desert ecosystem; snow leopard	—
Project Lion	2020	Asiatic lion (only at Gir, Gujarat)	—
Project Dolphin	2020	Gangetic river dolphin + marine dolphins	—
Project Cheetah	2022	World's first intercontinental cheetah translocation; Kuno NP (MP)	8 (Namibia) + 12 (South Africa)
Vulture Conservation	2006	Breeding centres; phase-out of diclofenac (the drug that decimated vultures)	—

✔ KEY FACT

Project Tiger (1973) — India's oldest and most famous conservation project. First reserve was **Bandipur (Karnataka)**. Madhya Pradesh has the most tiger reserves (9) and the most tigers (~785 as per 2022 census). Project Cheetah (2022) is the world's first intercontinental translocation of a large carnivore — cheetahs were extinct in India since 1952.

National Symbols of India

Category	Symbol
National animal	Royal Bengal Tiger (Panthera tigris tigris)
National bird	Indian Peafowl (Pavo cristatus)
National aquatic animal	Gangetic River Dolphin (Platanista gangetica) — declared 2009
National heritage animal	Indian Elephant (Elephas maximus indicus) — declared 2010

Category	Symbol
National flower	Lotus (Nelumbo nucifera)
National tree	Indian Banyan (Ficus benghalensis)
National fruit	Mango (Mangifera indica)
National river	Ganga — declared 2008
National reptile	King Cobra (Ophiophagus hannah)
National butterfly	Southern Birdwing (Troides minos)

PART Y — INTERNATIONAL CONVENTIONS (COMPLETE TABLE)

Year	Convention	Key outcome
1971	**Ramsar Convention**	Conservation of wetlands; India joined 1982; 85+ Indian sites
1972	**Stockholm Conference**	First UN env conference; UNEP formed (HQ Nairobi); World Environment Day = 5 June
1973	**CITES**	Trade in endangered species; 3 appendices; India joined 1976
1979	Bonn Convention (CMS)	Conservation of migratory species; India 1983
1985	**Vienna Convention**	Framework for ozone layer protection
1987	**Montreal Protocol**	Phase-out CFCs and other ODSs; India 1992; most successful env treaty
1989	Basel Convention	Trans-boundary movement of hazardous wastes; India 1992
1992	**Rio Earth Summit (UNCED)**	Agenda 21; UNFCCC, CBD signed; birth of modern env diplomacy
1992	**CBD**	Convention on Biological Diversity; conservation + sustainable use + benefit-sharing
1992	**UNFCCC**	UN Framework Convention on Climate Change; entered force 1994
1997	**Kyoto Protocol**	Binding GHG cuts on Annex-I (developed) countries only
2000	Cartagena Protocol	Biosafety of living modified organisms (LMOs); under CBD
2001	**Stockholm Convention (POPs)**	"Dirty dozen" persistent organic pollutants; India 2006
2010	**Nagoya Protocol**	Access + benefit sharing of genetic resources; under CBD
2010	Aichi Biodiversity Targets	20 targets under CBD for 2020
2013	Minamata Convention	Reduce mercury use and emissions; India 2018
2015	**Paris Agreement**	Limit warming to 1.5–2°C; voluntary NDCs by all parties
2015	SDGs	17 Sustainable Development Goals; 169 targets; UN GA
2016	**Kigali Amendment**	Phase-down HFCs (added to Montreal Protocol); India 2021
2021	Glasgow COP-26	India's PANCHAMRIT; coal phase-down
2022	Sharm el-Sheikh COP-27	Loss and Damage fund agreed in principle

Year	Convention	Key outcome
2022	**Kunming-Montreal GBF**	"30×30" — 30% of land + sea protected by 2030; CBD COP-15
2023	Dubai COP-28	First Global Stocktake; "transition away from fossil fuels"
2024	Baku COP-29	$300 billion/year climate finance by 2035
2025	Belém COP-30	Brazil host; first COP in the Amazon

PART Z — BIODIVERSITY (DETAILED)

Famous Environmental Movements

Movement	Year	Location	Leader(s)
Chipko movement	1973	Garhwal, Uttarakhand	Sundarlal Bahuguna, Chandi Prasad Bhatt
Silent Valley movement	1973–83	Kerala	Saved Silent Valley from a dam
Appiko movement	1983	Karnataka (Western Ghats)	Pandurang Hegde
Narmada Bachao Andolan	1985–	MP / Maharashtra / Gujarat	**Medha Patkar**, Baba Amte
Tehri Dam protest	1990s	Uttarakhand	Sundarlal Bahuguna

Global Environmental Days

Date	Day
2 February	World Wetlands Day
3 March	World Wildlife Day
21 March	International Day of Forests
22 March	World Water Day
22 April	Earth Day
22 May	International Day for Biological Diversity
5 June	**World Environment Day**
8 June	World Oceans Day
16 September	World Ozone Day
13 October	International Day for Disaster Risk Reduction

Key Acronyms

Acronym	Full Form
UNFCCC	UN Framework Convention on Climate Change

Acronym	Full Form
IPCC	Intergovernmental Panel on Climate Change
UNEP	UN Environment Programme
CBD	Convention on Biological Diversity
CITES	Convention on International Trade in Endangered Species
GEF	Global Environment Facility
IUCN	International Union for Conservation of Nature
COP	Conference of the Parties
NDC	Nationally Determined Contribution
REDD+	Reducing Emissions from Deforestation and forest Degradation
CPCB	Central Pollution Control Board
MoEFCC	Ministry of Environment, Forest & Climate Change
NGT	National Green Tribunal
NBA	National Biodiversity Authority
NTCA	National Tiger Conservation Authority
FSI	Forest Survey of India
WII	Wildlife Institute of India
CBDR	Common But Differentiated Responsibilities
GBF	Global Biodiversity Framework (Kunming-Montreal 2022)

APPENDIX A — Disaster Management (Brief)

- **Disaster Management Act 2005** — legal framework for disaster management in India.
- **NDMA** (National Disaster Management Authority) — chaired by the Prime Minister.
- **NDRF** (National Disaster Response Force) — operational since 2006; 16 battalions.
- **CDRI** (Coalition for Disaster Resilient Infrastructure) — launched by India in 2019; international initiative.

India's seismic zones: **Zones II to V** (4 zones by BIS). Zone V is the highest risk — covers J&K (Kashmir valley), parts of HP, Uttarakhand, all northeastern states, and Kutch region of Gujarat.

APPENDIX B — SDGs (Sustainable Development Goals)

17 Goals adopted by UN General Assembly in 2015, with 169 targets, to be achieved by 2030:

1. No Poverty | 2. Zero Hunger | 3. Good Health | 4. Quality Education | 5. Gender Equality | 6. Clean Water | 7. Clean Energy | 8. Decent Work | 9. Industry & Innovation | 10. Reduced Inequalities | 11. Sustainable Cities | 12. Responsible Consumption | 13. Climate Action | 14. Life Below Water | 15. Life on Land | 16. Peace & Justice | 17. Partnerships

SDG India Index is published annually by NITI Aayog.

APPENDIX C — 25-Question Mini-Mock

1. What year was the Stockholm Conference on Human Environment held?
2. Which city hosts UNEP headquarters?
3. When is World Environment Day celebrated?
4. What is the Ramsar Convention about and in what year was it signed?
5. India's first Ramsar site(s)?
6. How many biodiversity hotspots does India have?
7. When was the Bhopal gas tragedy and what chemical leaked?
8. What year was India's Environment Protection Act passed?
9. What year was the Wildlife Protection Act passed?
10. How many schedules does the WLPA now have (after 2022)?
11. When was NTCA established?
12. What year was Project Tiger launched?
13. What is India's national aquatic animal?
14. What is India's national heritage animal?
15. What year was the Montreal Protocol signed?
16. What year was the Kyoto Protocol signed?
17. What year was the Paris Agreement signed?
18. What is India's net-zero target year?
19. What is the "30×30" target and which agreement does it come from?
20. What does PANCHAMRIT refer to?
21. Which Act created the National Green Tribunal?
22. What is India's current forest cover as a percentage of its area?
23. How many SDGs are there and by when must they be achieved?
24. India's largest biosphere reserve by area is in which state?
25. Name the only floating National Park in India and its state.

Answers: 1. 1972 2. Nairobi, Kenya 3. 5 June 4. Conservation of wetlands; 1971 (at Ramsar, Iran) 5. Chilika (Odisha) and Keoladeo Ghana (Rajasthan) — both designated 1981 6. 4 (Himalayas, Indo-Burma, Western Ghats+Sri Lanka, Sundaland) 7. 2-3 December 1984; Methyl Isocyanate (MIC) from Union Carbide 8. 1986 9. 1972 10. 4 schedules 11. 2005 12. 1973 13. Gangetic River Dolphin (declared 2009) 14. Indian Elephant (declared 2010) 15. 1987 16. 1997 17. 2015 18. 2070 (announced at COP-26, Glasgow, 2021) 19. Protecting 30% of Earth's land and sea by 2030; Kunming-Montreal Global Biodiversity Framework (2022) 20. India's 5 climate commitments made at Glasgow COP-26 (2021) 21. National Green Tribunal Act, 2010 22. 25.17% (forest + tree cover) 23. 17 goals; by 2030 24. Gujarat (Kachchh Biosphere Reserve is the largest) 25. Keibul Lamjao National Park; Manipur (on Loktak Lake)

APPENDIX D — Trap Recognition Card

Trap	Common mistake	Correct answer
EPA year	1984 (Bhopal year)	1986 — passed 2 years after Bhopal
Stockholm — which one?	Mixing 1972 conf and 2001 POPs convention	1972 = Conference on Human Env; 2001 = POPs Convention
Vienna vs Montreal	Same thing	Different: Vienna 1985 = framework; Montreal 1987 = actual CFC phase-out
Kyoto binds whom?	All countries	Only Annex-I (developed) countries
India's first Ramsar site	Chilika only	Both Chilika AND Keoladeo (1981)
WLPA schedules	6	4 (after 2022 amendment)
India joined Montreal Protocol	1987	1992 — India joined 5 years after the protocol was signed
India's net-zero target	2050 or 2060	2070
First NP of India	Jim Corbett established directly	Hailey NP (1936) → renamed Jim Corbett
Only Asiatic lions	Multiple locations	Gir NP, Gujarat — the ONLY place in the world
Project Tiger first reserve	Corbett	Bandipur, Karnataka (1973)
Biosphere Reserves on UNESCO list	All 18	Only 12 of India's 18 BRs are on UNESCO MAB World Network
Floating NP	Some river-based NP	Keibul Lamjao (Manipur, Loktak Lake) is the ONLY one
Largest NP in India	Sundarbans or Kaziranga	Hemis NP (Ladakh, ~4,400 km²)